The LNM Institute of Information Technology, Jaipur
Software Development Laboratory
Course Code: CS314
Lab Assignment 1: Solution

1. Project Identification

Item	Details
Project Name	LNMIIT Degree Audit System
Project Type	Web-based system that manages academic requirements for graduation
Team ID	T030
Team Members	Hatwar Manas Vinod, Dhruv Gupta, Aradhya Raina, Kota Sai Vignesh
Roll Numbers	24UCC176, 24UCC177, 24UCC178, 24UCC175
Development Model	Agile + Incremental Development
Primary Users	Students, Registrar, Academic Administrators
Development Team	Student software development team
Course	Software Engineering
Project Objective	To develop a platform that keeps records of students' academic progress and automatically evaluates its completion status and requirements to be eligible for graduation degree attainment
Expected Outcome	A properly working, tested and documented software system developed according to requirements

2. Project Background

Current Problem

Academic progress tracking at LNMIIT is currently managed through disconnected mechanisms such as manual transcript reviews, static PDF catalogs, spreadsheets and paper-based records. This can make it difficult for students to understand their degree completion status, plan future courses, explore program changes, and prepare for graduation. Academic advisors lack real-time visibility into student progress, and the Registrar office must manually verify graduation requirements for each student during peak graduation periods.

Proposed Solution

The LNMIIT Degree Audit System will provide a centralized platform for managing the complete academic progress lifecycle—from automated transcript analysis and requirement validation to what-if scenario exploration, course planning, and official graduation clearance reporting.

3. Project Vision

The LNMIIT Degree Audit System aims to provide a centralized, user-friendly and reliable platform for tracking academic progress, validating degree requirements, exploring program changes, and planning course sequences to ensure timely graduation.

4. Problem Statement

LNMIIT conducts undergraduate and graduate academic programs across four departments involving diverse degree requirements, major/minor combinations, transfer credits, and catalog year variations. Existing progress tracking practices rely on multiple disconnected tools and manual processes, resulting in difficulties in requirement validation, course planning, what-if analysis, graduation clearance, academic advising efficiency, and early identification of at-risk students. The LNMIIT Degree Audit System is proposed as an integrated software solution to streamline these activities through a single automated platform.

5. Project Objectives

The initial objectives should be:

1. provide a centralized platform for tracking students' academic progress
2. Provide role-based access control
3. Allow authorized administrators to define, update, and maintain degree requirements, course categories, and credit thresholds.
4. Automate degree requirement comparison
5. Automate credit and deficiency analysis b
6. Provide students with secure access to their real-time degree requirement completion status
7. Generate degree audit reports s
8. Maintain data integrity and security
9. To support all LNMIIT departments and multiple catalog years.
10. To develop the system incrementally using software engineering practices.

6. Stakeholders

Stakeholder	Interest / Role
Students (UG & PG)	View degree progress, plan courses, explore major/minor options, check graduation eligibility
Faculty Advisors	Monitor advisee academic progress, provide course planning guidance, approve semester registrations
HODs	Review departmental graduation requirements, verify BTP/thesis completion, approve degree conferral
Registrar Office	Final graduation clearance authority, maintain official catalog requirements, issue degree certificates
Admission Cell	Evaluate and record transfer credits for lateral entry, approve course substitutions and equivalencies
Dean of Academic Affairs	Oversee institutional academic standards, review policy compliance, handle exception requests
System Development Team	Design, develop, test and maintain the LNMIIT Degree Audit System

7. Initial Scope

The Degree Audit System will initially cover:

• User authentication/login
• Role-based access
• Semester grades and course information imported by administrators
• Automatic comparison of completed courses/credits against curriculum requirements
• Progress dashboard
• Requirement breakdown for different degrees
• Multi-catalog year support
• Prerequisite verification
• Audit report generation (view/download)
• Advisor monitoring dashboard
• At-risk student alerts
• Basic analytics/reports

Initially Out of Scope

To keep the project manageable, we should initially exclude things such as:

- What-if degree planning simulators or AI-based course recommendations
- Mobile native application (web-based only initially)
- Support for multiple universities or cross-institution transfer credit evaluation
- Automated official transcript generation for external use
- Complete ERP functionality
- Real-time faculty/grade-entry integration (grades assumed to be entered via existing academic records before import)

8. Initial Assumptions

- Academic catalogs accurately reflect official degree requirements.
- Student transcript data is available for import.
- Program requirements remain relatively stable within an academic year
- The system will initially be developed as a web application.
- The system will use a centralized database.
- LNMIIT IT will provide hosting infrastructure and technical support.
- System-generated audit reports will be accepted as official by Registrar.
- Students may choose catalog from admission year forward.

9. Initial Constraints

- Limited development time within the semester.
- Limited access to real institutional records
- Degree requirements may differ across academic programs.
- The system will initially be developed and tested using a limited set of degree programs and sample
- Requirement changes may occur during the development process.
- Limited testing with real students, faculty, and academic administrators.

- Student academic records and degree-related information need to be appropriately protected.
- Requirement changes may occur during the development process.
- Integration with existing institutional academic/ERP systems may not be available during development.
- Limited infrastructure and resources available for development and testing.
- The project is developed by a student team, limiting the time and resources available for implementing a large-scale institutional system.

10. Initial Success Criteria

- different user roles can access appropriate functionality.
- student transcripts are accurately imported and validated.
- students can view progress dashboards showing completed and remaining requirements.
- what-if analysis allows exploring major/minor changes.
- The system supports all departments and multiple catalog years.
- transfer credits and course substitutions are tracked correctly.
- official audit reports are generated and accepted by Registrar office.
- advisors can monitor advisees and identify at-risk students.
- important system functions are tested.
- defects are documented and resolved.
- The project is maintained in a controlled Git/GitHub repository.
- The final system is accompanied by appropriate software engineering documentation.

AI USAGE POLICY – STUDENT DECLARATION

We declare that:

• We have disclosed the use of AI tools, if any, during preparation of this assignment.

• AI-generated content, if used, has been reviewed, verified, edited and adapted by the team.

• We understand the project sufficiently to explain and defend all submitted content during a viva/discussion.

• We have not submitted AI-generated content blindly or without verification.

• We have not used AI to misrepresent work performed by the team.

• The final document reflects the team's own project understanding and decisions.

AI Tool(s) Used: Claude

Purpose of AI Use:

- To review the project-definition document for completeness, consistency, and alignment with the assignment requirements
- Help in Search for Resources to understand, plan and decide on the project vision, problem statement, objectives, scope, and stakeholder descriptions of the degree Audit System.

- To identify areas that may require clarification or improvement.
- To Check for Grammatical Mistakes

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Date:
30 AUGUST 2026

Team Declaration:
We confirm that the information provided above is accurate and that we accept responsibility for the submitted work.

Team Representative Signature/Confirmation:
DHRUV GUPTA